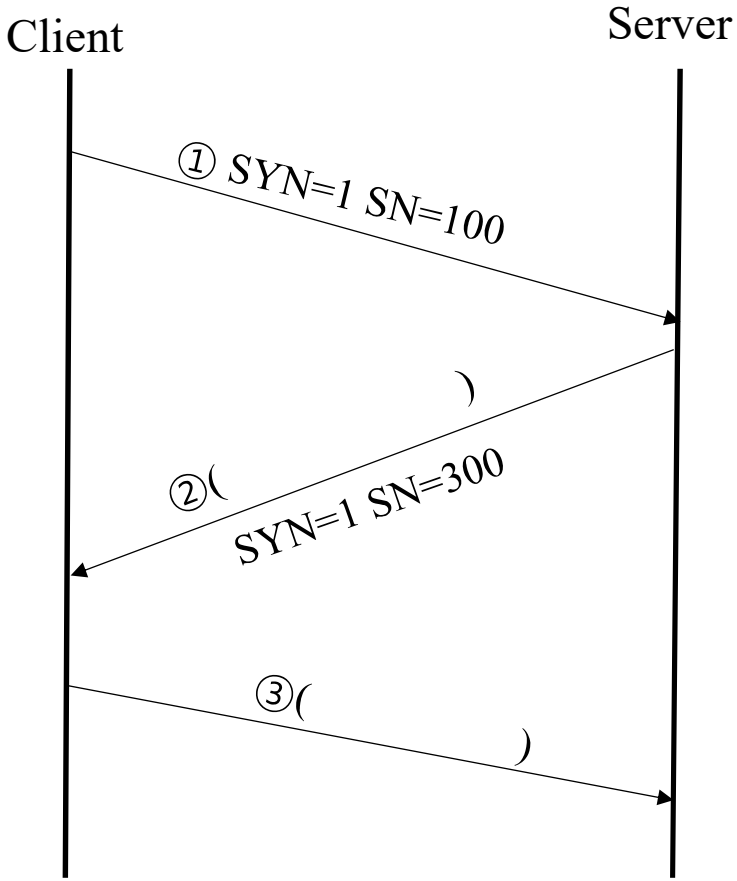
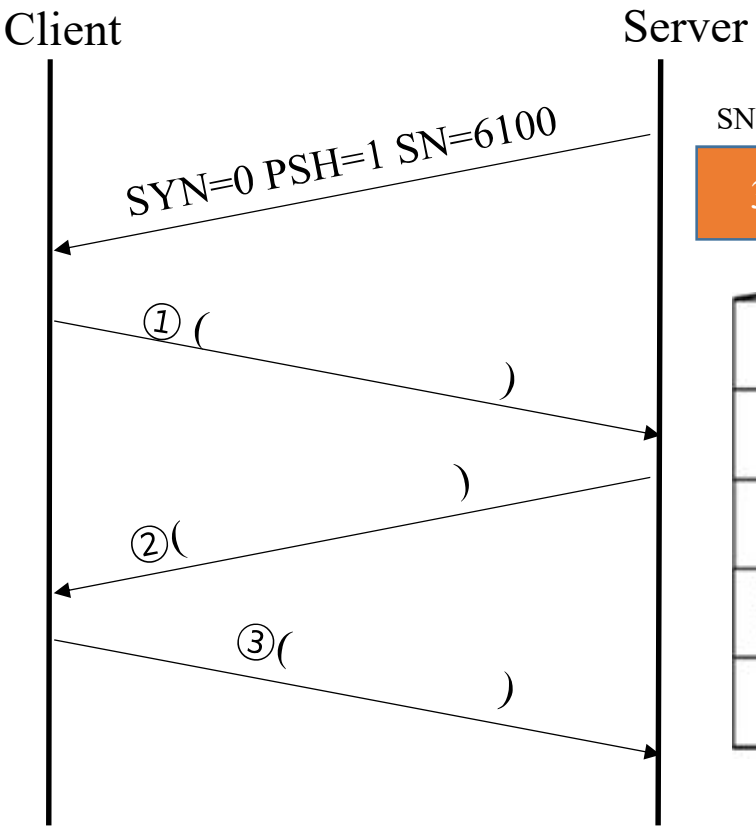
TCP Process

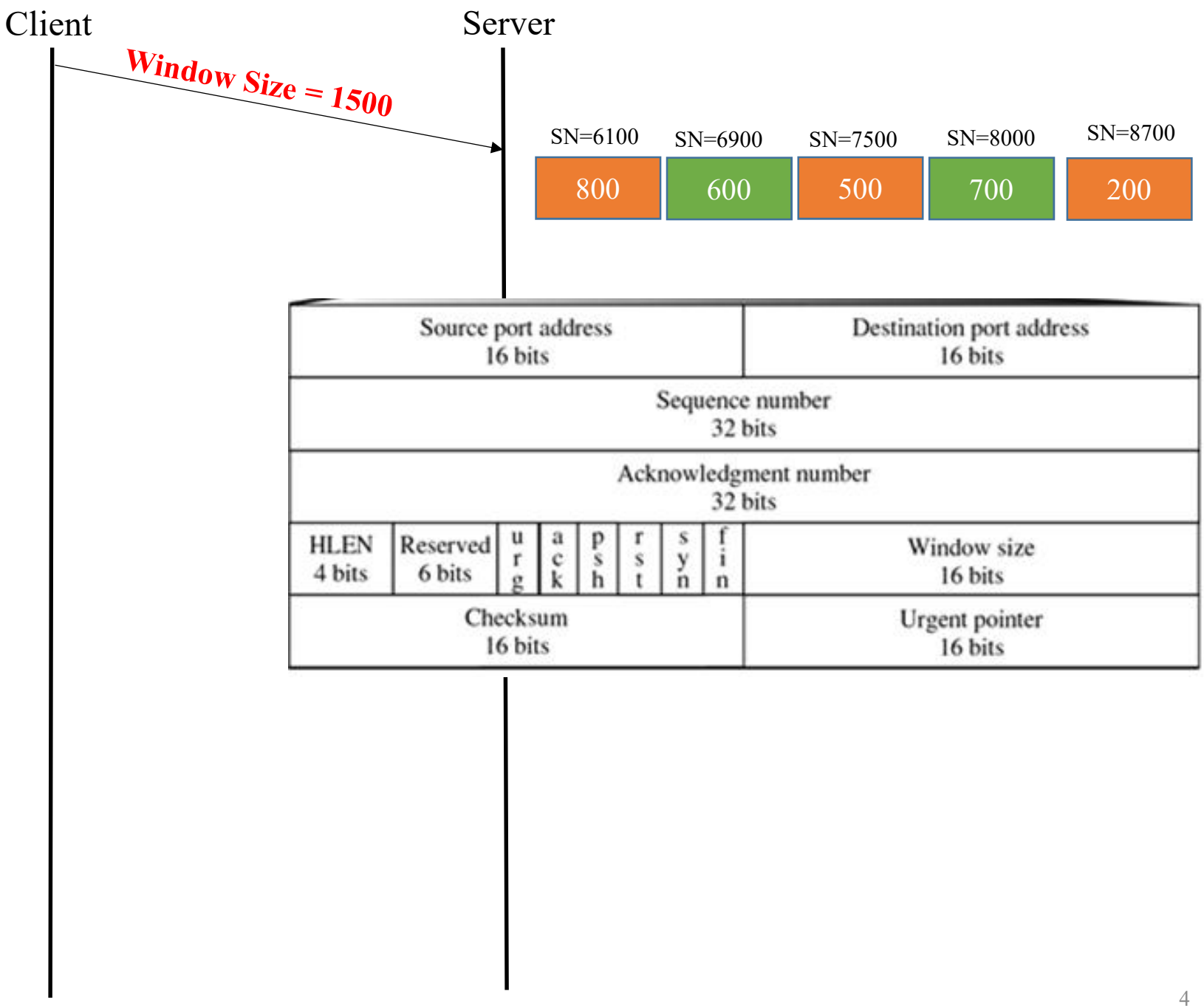


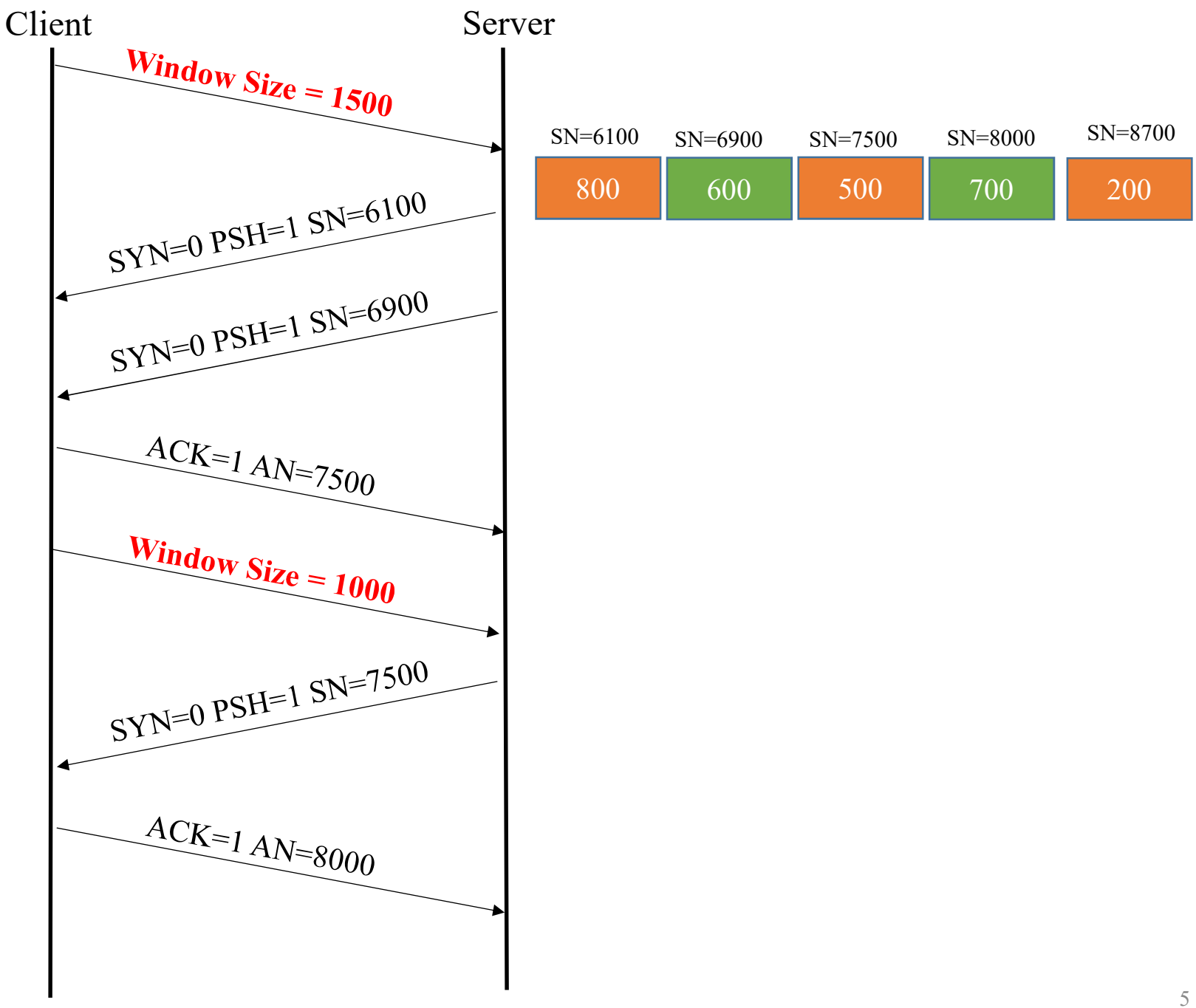
Source port address 16 bits								Destination port address 16 bits
Sequence number 32 bits								
Acknowledgment number 32 bits								
HLEN 4 bits	Reserved 6 bits	urg	ack	psh	rst	syn	fin	Window size 16 bits
Checksum 16 bits								Urgent pointer 16 bits

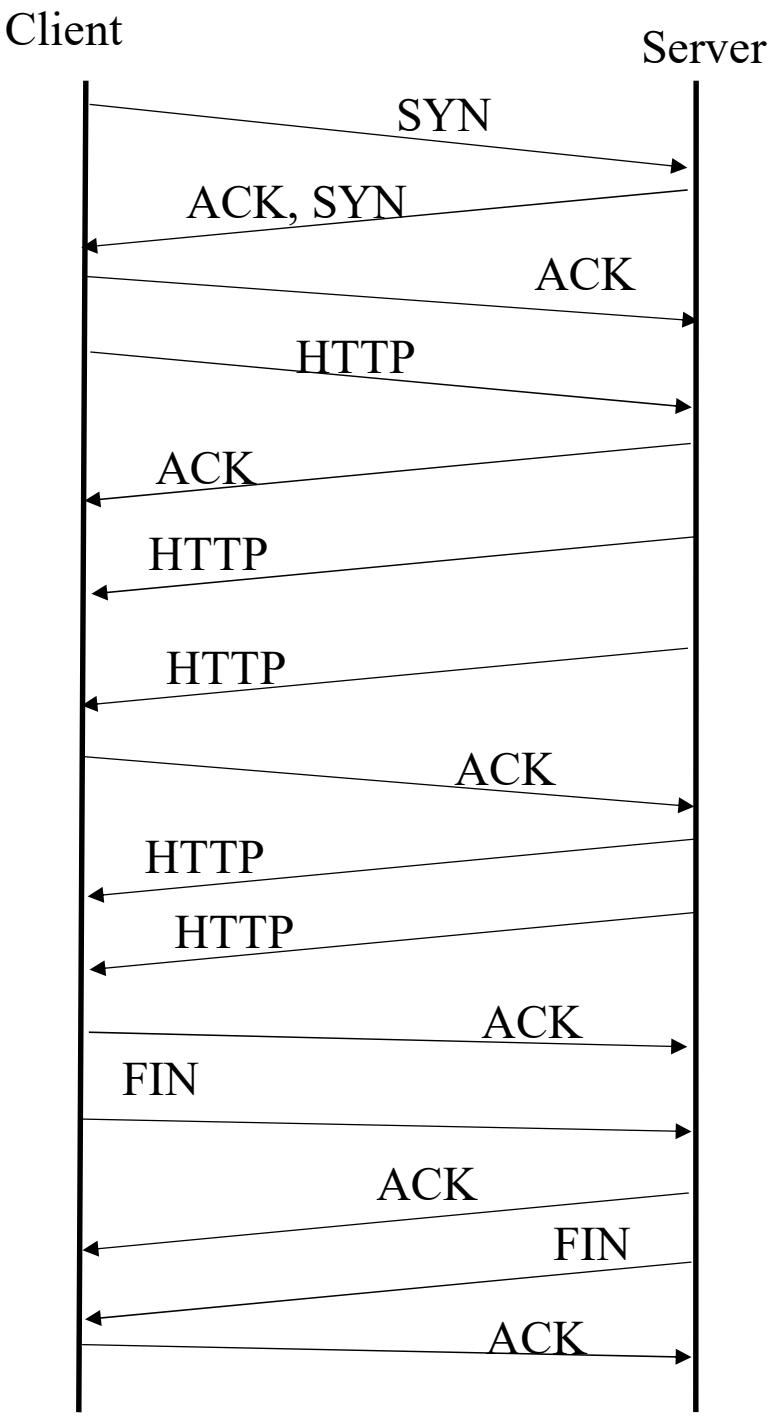
Source port address 16 bits								Destination port address 16 bits
Sequence number 32 bits								
Acknowledgment number 32 bits								
HLEN 4 bits	Reserved 6 bits	urg	ack	psh	rst	syn	fin	Window size 16 bits
Checksum 16 bits								Urgent pointer 16 bits



Source port address 16 bits								Destination port address 16 bits
Sequence number 32 bits								
Acknowledgment number 32 bits								
HLEN 4 bits	Reserved 6 bits	urg	ack	psh	rst	syn	fin	Window size 16 bits
Checksum 16 bits								Urgent pointer 16 bits


Client
Server
Window Size = 1500
SN=6100
SN=6900
SN=7500
SN=8000
SN=8700
800
600
500
700
200
Source port address
16 bits
Destination port address
16 bits
Sequence number
32 bits
Acknowledgment number
32 bits
HLEN
4 bits
Reserved
6 bits
urg
ack
psh
rst
syn
fin
Window size
16 bits
Checksum
16 bits
Urgent pointer
16 bits


Client
Server
Window Size = 1500
SN=6100
SN=6900
SN=7500
SN=8000
SN=8700
800
600
500
700
200
SYN=0 PSH=1 SN=6100
SYN=0 PSH=1 SN=6900
ACK=1 AN=7500
Window Size = 1000
SYN=0 PSH=1 SN=7500
ACK=1 AN=8000


Client
Server
SYN
ACK, SYN
ACK
HTTP
ACK
HTTP
HTTP
ACK
HTTP
HTTP
ACK
FIN
ACK
FIN
ACK

ARP

ARP(Address Resolution Protocol) 패킷

- IP 주소에 대응 되는 MAC 주소를 조회 변환해 주는 서비스
- ARP 패킷 종류

❶ ARP request 패킷

- 송신지가 수신지의 MAC 주소를 조회하기 위해 보내는 질의 패킷
- 브로드캐스트 방식으로 운영

❷ ARP reply 패킷

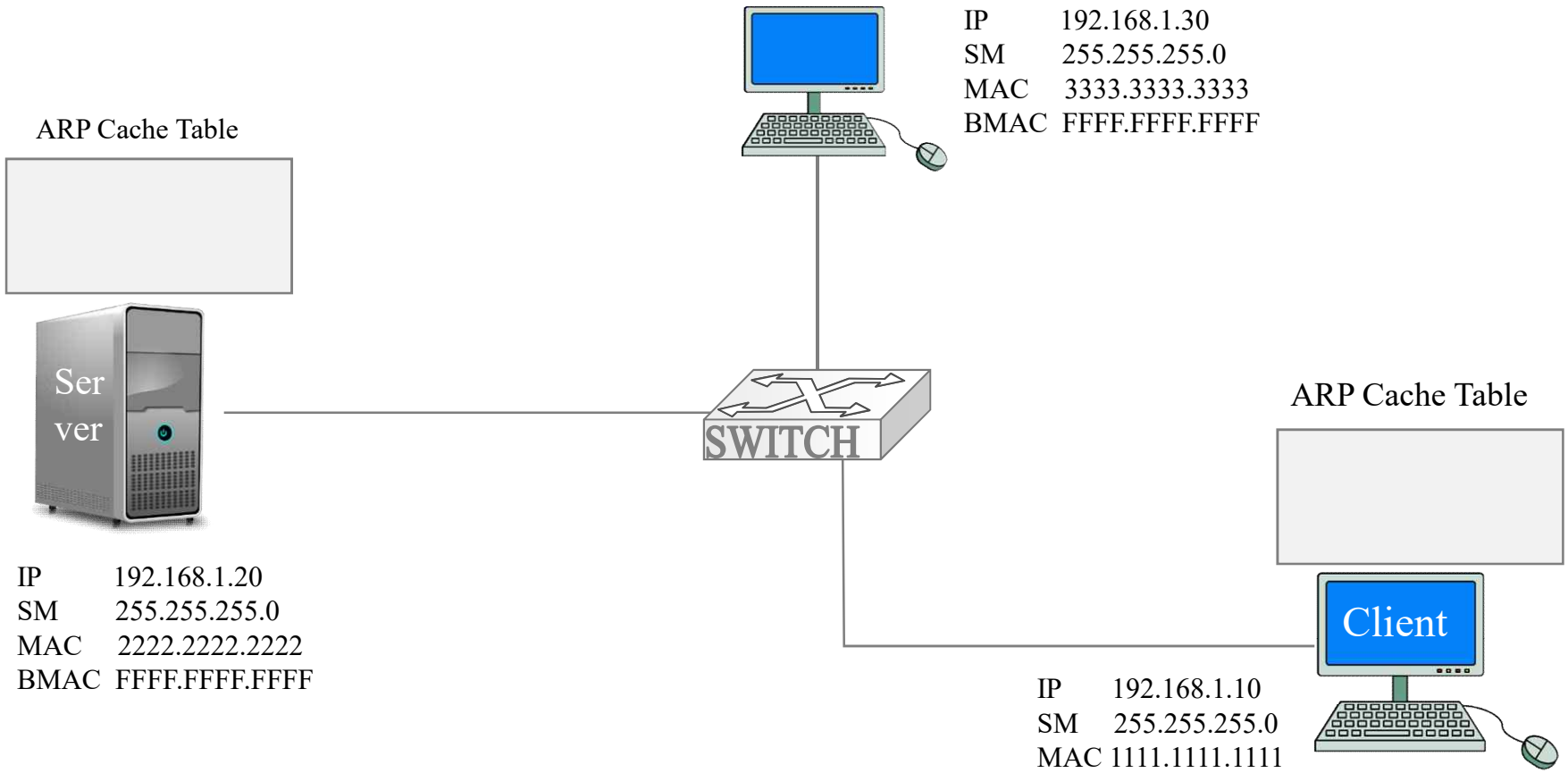
- ARP request에 대해 응답 패킷
- 유니캐스트 방식으로 운영

ARP 캐시 테이블(Cache Table)

- IP주소와 MAC 주소의 대응 관계를 저장한 테이블
- ARP 캐쉬 테이블 확인 명령어 : arp -a

```
C:₩>arp -a

인터페이스: 192.168.35.131 --- 0x4
  인터넷 주소           물리적 주소           유형
  192.168.35.1          00-23-aa-83-11-69     동적
  192.168.35.115        38-8c-50-9a-9b-b1     동적
  192.168.35.211        04-b4-29-bf-07-09     동적
  192.168.35.255        ff-ff-ff-ff-ff-ff     정적
  224.0.0.22            01-00-5e-00-00-16     정적
  224.0.0.251           01-00-5e-00-00-fb     정적
  224.0.0.252           01-00-5e-00-00-fc     정적
  239.255.255.250       01-00-5e-7f-ff-fa     정적
  255.255.255.255       ff-ff-ff-ff-ff-ff     정적
```



ARP Request

수MAC	송MAC	Protocol	ARP Header	
FFFF.FFFF.FFFF	1111. 222.3333	0x0806	1111.1111.1111 /192.168.1.10 **0000.0000.0000/192.168.1.20**	송신 수신

수MAC	송MAC	송IP	수IP	송Port	수Port	전송데이타
????	1111. 1111.1111	192.168.1.10	**192.168.1.20**	50030	80	Get /www.test.com

	ARP Header	Protocol	송MAC	수MAC
송신 수신	2222.2222.2222 /192.168.1.20 1111.1111.1111 /192.168.1.10	0x0806	2222.2222.2222	1111.1111.1111

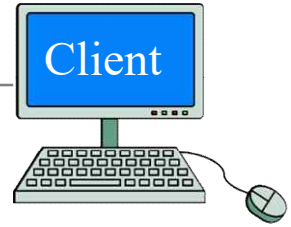
ARP Cache Table

192.168.1.10	**1111.1111.1111**

IP 192.168.1.20
SM 255.255.255.0
MAC 2222.2222.2222
BMAC FFFF.FFFF.FFFF

ARP Cache Table

192.168.1.20	**2222.2222.2222**



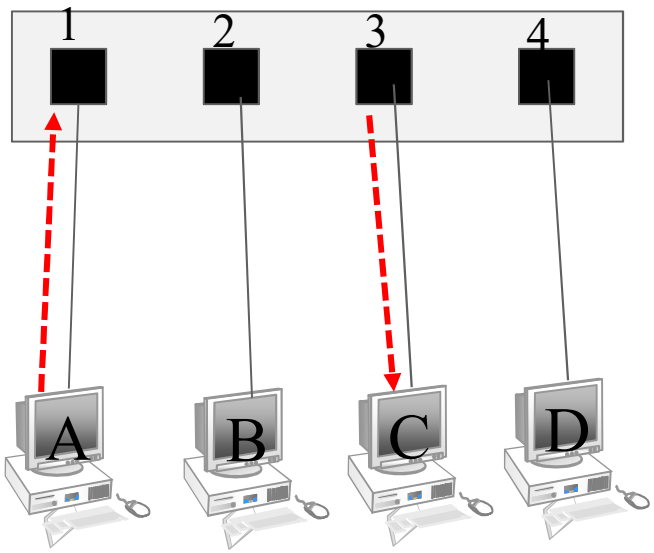
IP 192.168.1.10
SM 255.255.255.0
MAC1111.1111.1111

수MAC	송MAC	송IP	수IP	송Port	수Port	전송데이타
222.2222.2222	1111. 1111.1111	192.168.1.10	**192.168.1.20**	50030	80	Get /www.test.com

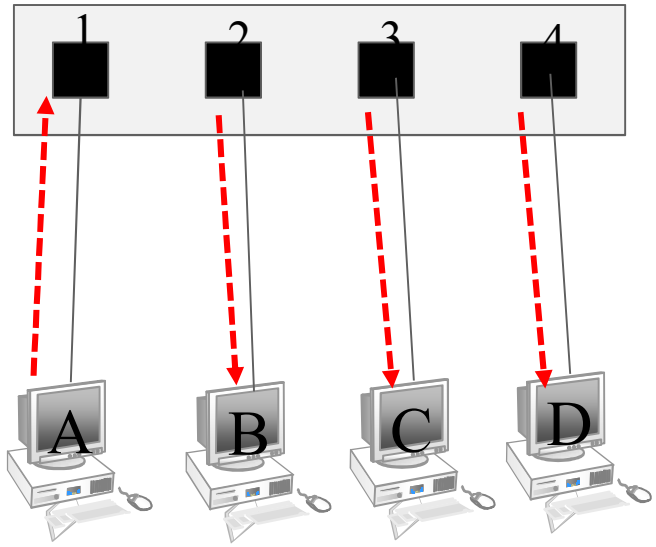
장비 특성

Forwarding과 Flooding

- Forwarding(포워딩)
  - 하나의 송신지 포트에서 하나의 수신지 포트로 트래픽 전송
- Flooding(플러딩)
  - 송신지 포트를 제외한 나머지 포트들로 트래픽 전송



Forwarding(포워딩)



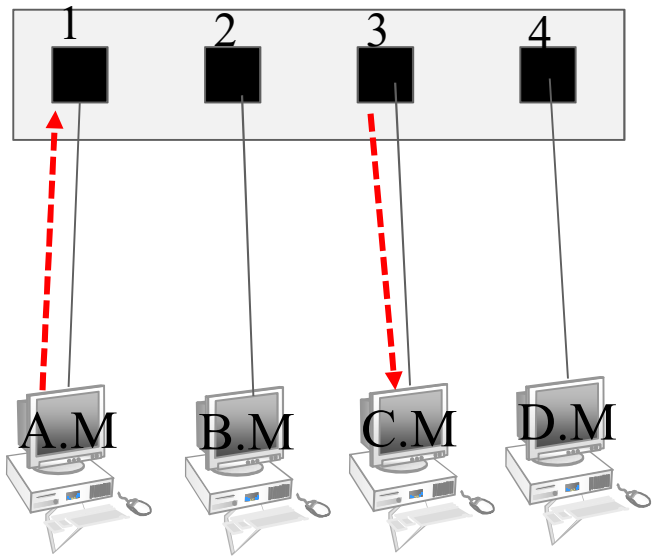
Flooding(플러딩)

계층별 장비- 2계층 장비

MAC Address Table

목적지	출구번호
A.M	1
B.M	2
C.M	3

❶

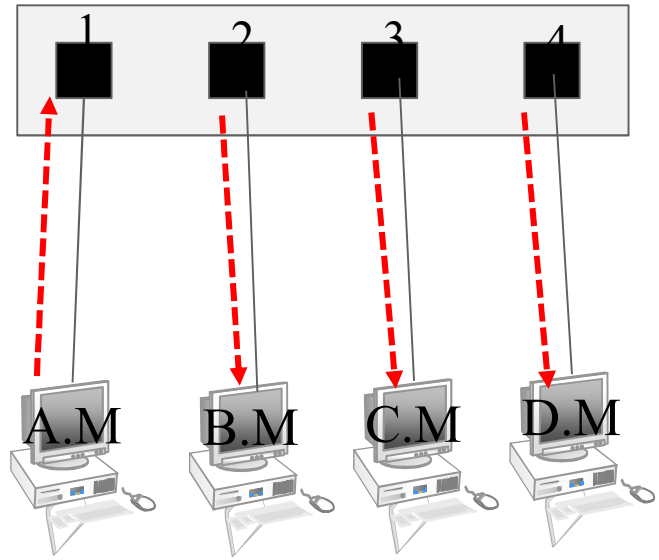


송신지MAC	수신지MAC
A.M	C.M

MAC Address Table

목적지	출구번호
A.M	1
B.M	2
C.M	3

❷

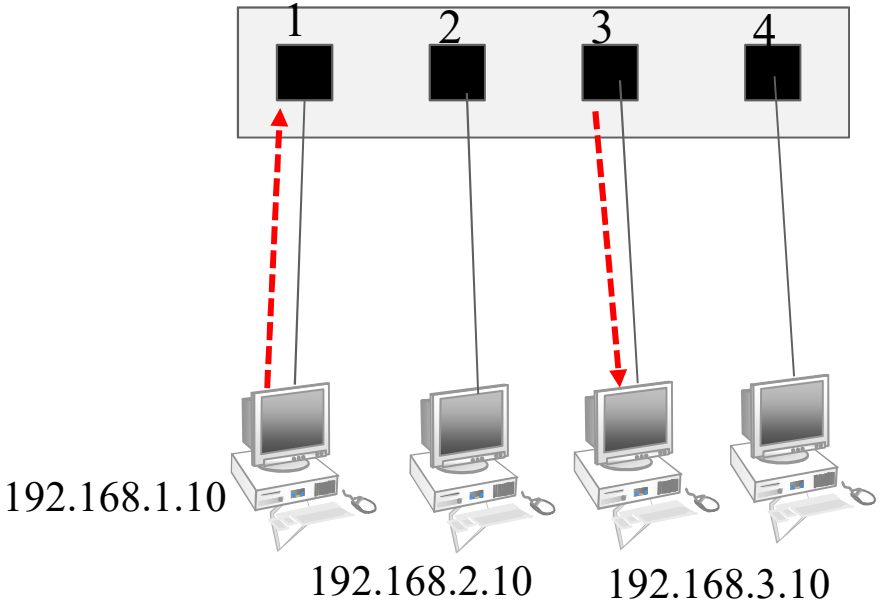


송신지MAC	수신지MAC
A.M	D.M

계층별 장비- 3계층 장비

Routing Table

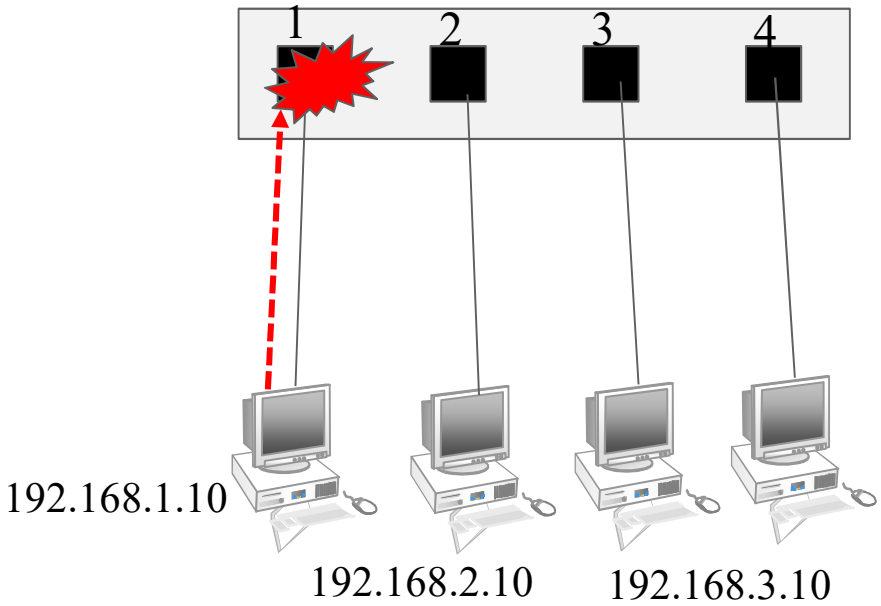
목적지	출구번호
192.168.1.0	1
192.168.2.0	2
192.168.3.0	3



192.168.1.10	192.168.3.10
송신지IP	수신지IP

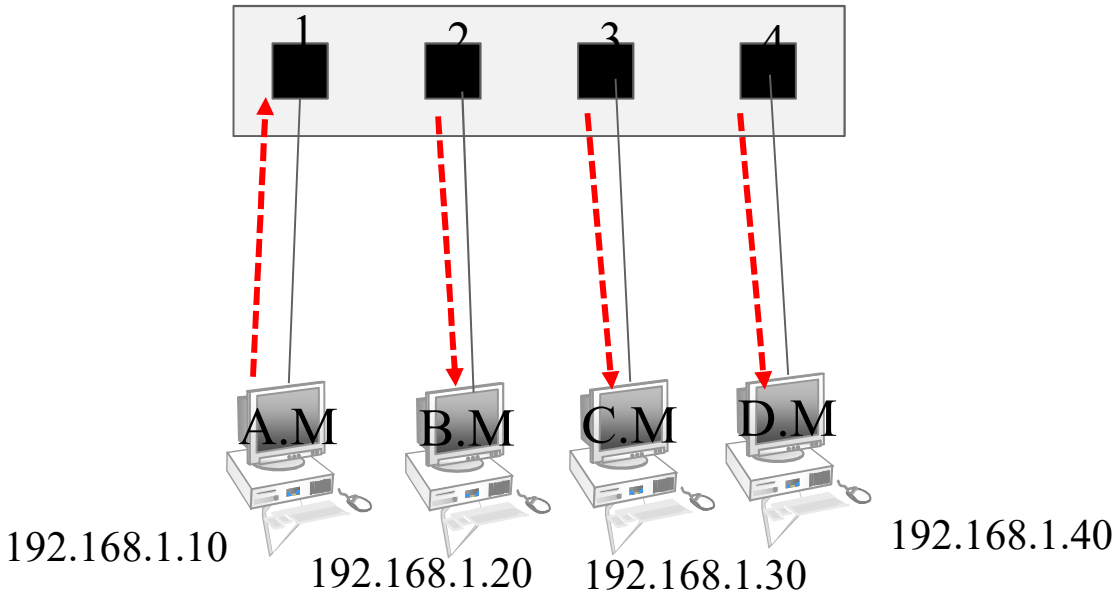
Routing Table

목적지	출구번호
192.168.1.0	1
192.168.2.0	2
192.168.3.0	3

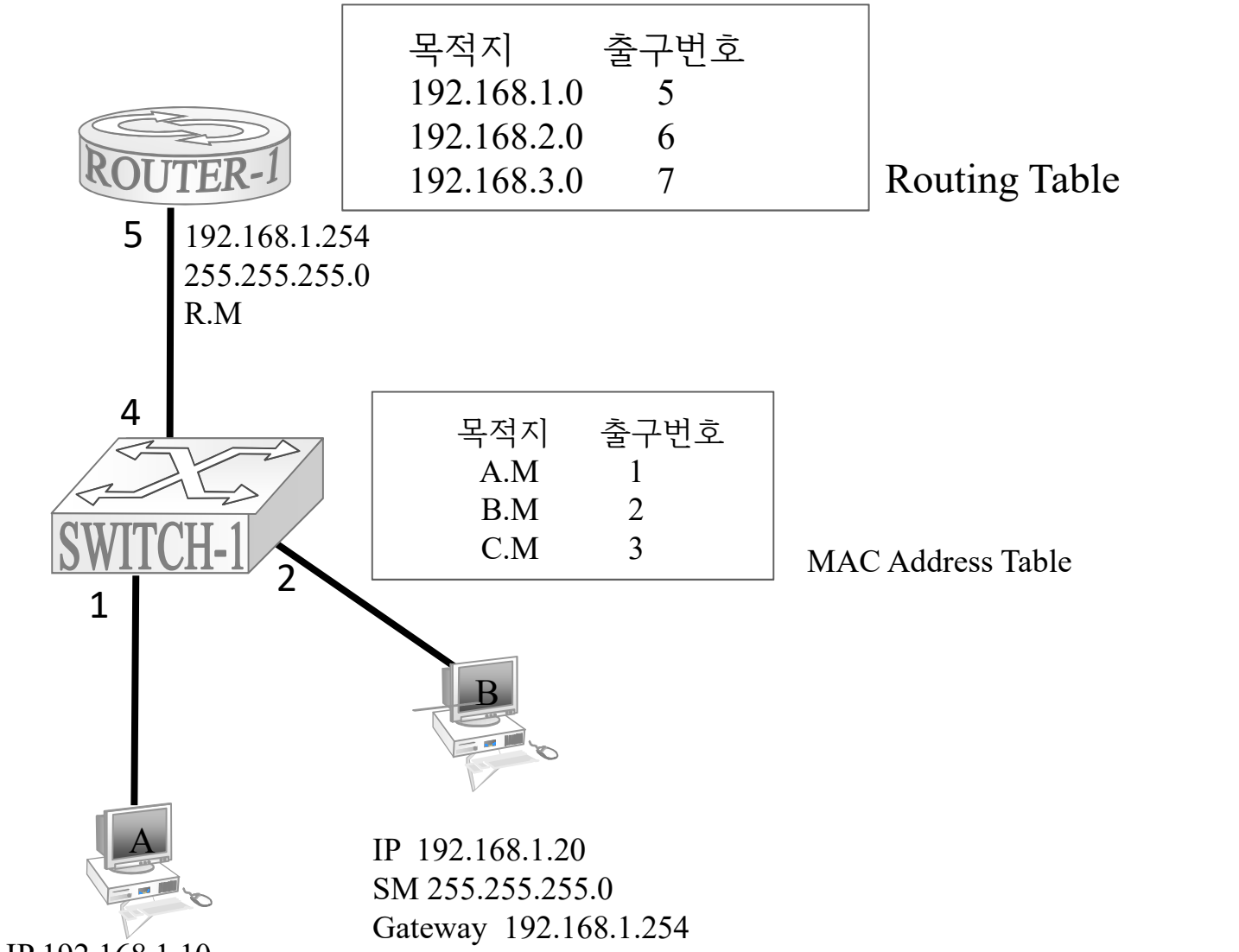


192.168.1.10	192.168.4.10
송신지IP	수신지IP

계층별 장비- 1계층 장비



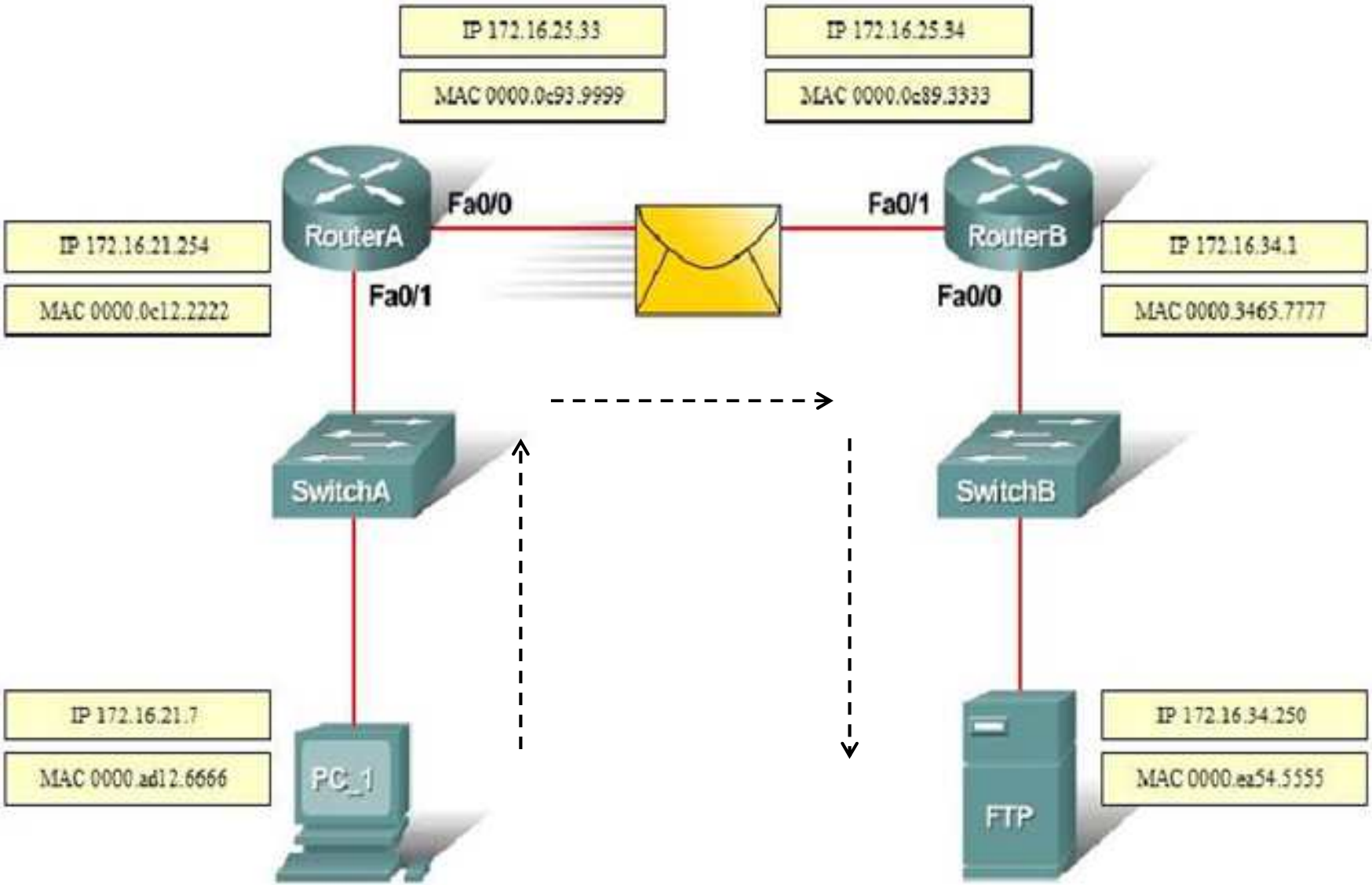
192.168.1.10	192.168.4.10	A.M	D.M
송신지IP	수신지IP	송MAC	수MAC



192.168.1.10	192.168.5.10	A.M	R.M
송신지IP	**수신지IP**	송신지MAC	수신지MAC

Media Translation

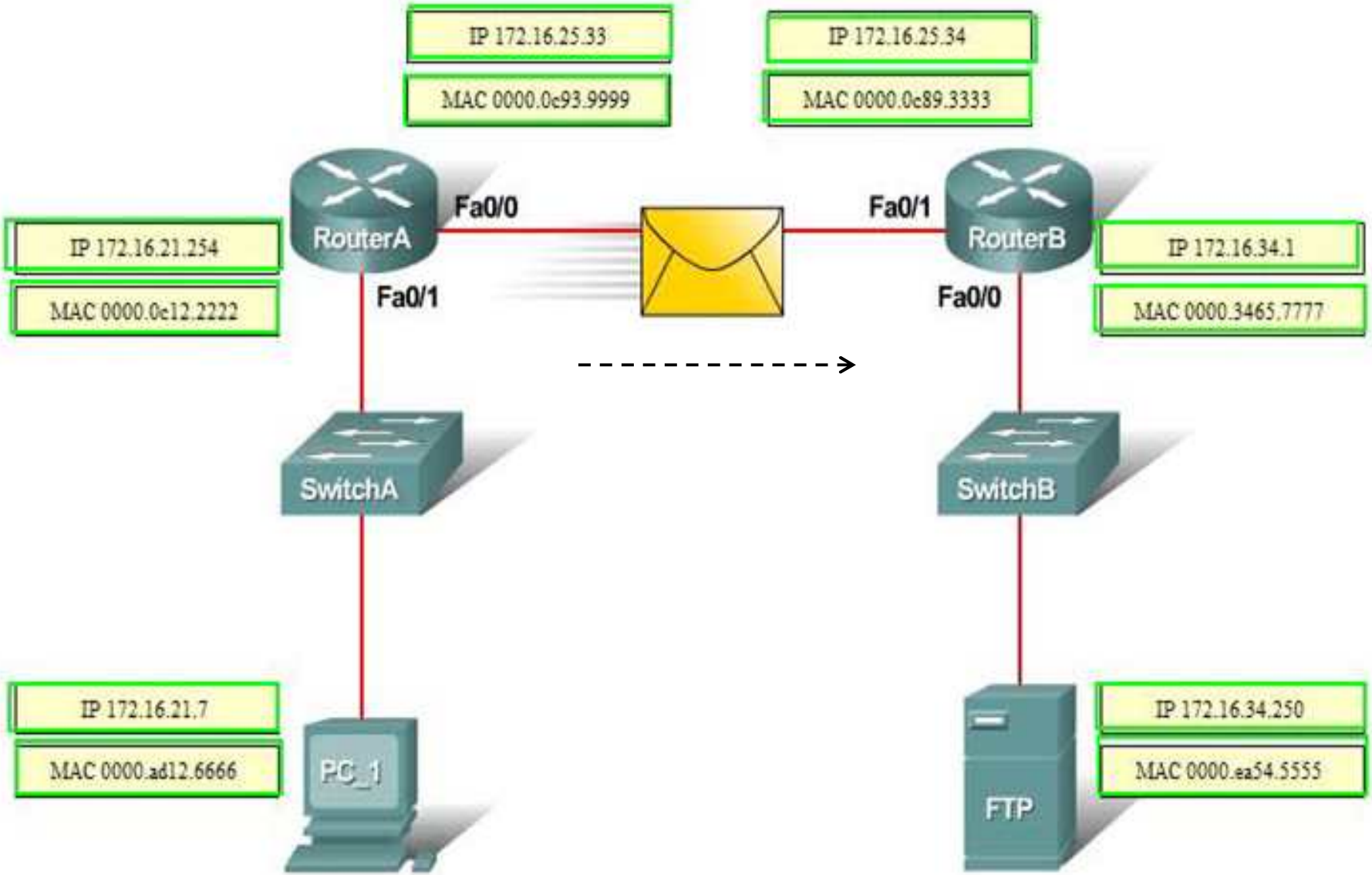
- 라우터에서 처리
- 패킷이 출발지에서 목적지까지 가는 동안 라우터를 거칠 때마다 L2헤더 (프레임 헤더)를 변경
- 2 계층 스위칭 환경에 따라 L2주소 정보 변환
- IP 주소 변환은 없음



Source MAC	Destination MAC	Source IP Address	Destination IP Address
Target	Target	Target	Target

Media Translation

	송 IP	수IP	송MAC	수MAC
PC-1 ---> RA	172.16.21.7	172.16.34.250	6666	2222
RA ---> RB	172.16.21.7	172.16.34.250	9999	3333
RB ---> FTP	172.16.21.7	172.16.34.250	7777	5555



Source MAC	Destination MAC	Source IP Address	Destination IP Address
MAC 0000.0c93.9999	MAC 0000.0c89.3333	IP 172.16.21.7	IP 172.16.34.250